

# Sistema de Gestion Club de Natacion - G2

## Estructura final del proyecto MVC en PHP

Modelo unificado: **users + profiles** (sin tablas swimmers ni coaches).

|                                                |                                                        |                                                            |                                         |
|------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|-----------------------------------------|
| <b>Backend</b><br>PHP 8 + PDO<br>MVC artesanal | <b>Frontend</b><br>Vistas .view.php<br>HTML / CSS / JS | <b>Base de datos</b><br>MariaDB 10.4<br>phpMyAdmin (XAMPP) | <b>Roles</b><br>Admin / Coach / Swimmer |
|------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------|-----------------------------------------|

## Documento de referencia

Versión 1.0 - Mayo 2026 - Grupo 2



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## 1 Resumen ejecutivo y decisiones de diseño

Este documento define la estructura final del Trabajo Practico, partiendo del código actual y aplicando una **simplificación del modelo de datos**: se eliminan las tablas **swimmers** y **coaches** y todos los datos personales (de cualquier rol) se concentran en **profiles**. El rol del usuario se mantiene en **users.role\_id**.

### Beneficios de la unificación

|                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| Una sola fuente de verdad | Los datos personales (nombre, telefono, foto) viven en una unica tabla. |
| Menos joins y modelos     | Desaparecen los modelos Swimmer y Coach; se reemplazan por Profile.     |
| Cambio de rol simple      | Promover un swimmer a coach es modificar role_id, sin migrar registros. |
| Especialidad opcional     | El campo specialty queda en profiles y se completa solo para coaches.   |

### Decisiones clave

|                            |                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------|
| FK de lessons              | coach_id referencia users.id (donde role_id = 2). No referencia a profiles.                     |
| FK de bookings             | swimmer_id referencia users.id (donde role_id = 3).                                             |
| Soft delete                | Se mantiene deleted_at en users y profiles. En bookings se usa el enum status.                  |
| Cancelacion booking        | status pasa a 'Cancelled'; no se borra el registro.                                             |
| Datos personales en sesion | Login carga users + profiles en \$_SESSION para que cada controller pueda usarlos sin re-query. |

## 2 Modelo de datos unificado

Esquema final con seis tablas (eliminadas swimmers y coaches).

| users (auth)      |                    |                 |
|-------------------|--------------------|-----------------|
| <b>id</b>         | int PK             | AI              |
| <b>email</b>      | varchar(100)<br>UQ | login           |
| <b>password</b>   | varchar(255)       | bcrypt          |
| <b>role_id</b>    | FK roles.id        | rol del usuario |
| <b>created_at</b> | timestamp          |                 |
| <b>updated_at</b> | timestamp          |                 |
| <b>deleted_at</b> | timestamp<br>NULL  | soft delete     |

| roles            |             |                                           |
|------------------|-------------|-------------------------------------------|
| <b>id</b>        | int PK      | 1=Administrator,<br>2=Coach,<br>3=Swimmer |
| <b>role_name</b> | varchar(50) | nombre del rol                            |

| bookings (inscripciones) |                            |                                   |
|--------------------------|----------------------------|-----------------------------------|
| <b>id</b>                | int PK                     | AI                                |
| <b>swimmer_id</b>        | FK users.id                | user con role_id = 3              |
| <b>lesson_id</b>         | FK lessons.id              |                                   |
| <b>status</b>            | enum                       | Confirmed  <br>Cancelled          |
| <b>created_at</b>        | timestamp                  |                                   |
| <b>UQ</b>                | (swimmer_id,<br>lesson_id) | evita inscripciones<br>duplicadas |

| profiles (datos personales unificados) |                      |                                      |
|----------------------------------------|----------------------|--------------------------------------|
| <b>id</b>                              | int PK               | AI                                   |
| <b>user_id</b>                         | FK users.id<br>UQ    | 1:1 con users (ON<br>DELETE CASCADE) |
| <b>first_name</b>                      | varchar(100)         | obligatorio                          |
| <b>last_name</b>                       | varchar(100)         | obligatorio                          |
| <b>phone</b>                           | varchar(20)          |                                      |
| <b>birth_date</b>                      | date NULL            | edad / mayoría                       |
| <b>specialty</b>                       | varchar(100)<br>NULL | solo coaches                         |
| <b>profile_image</b>                   | varchar(255)         | default-profile.png                  |
| <b>created_at</b>                      | timestamp            |                                      |
| <b>updated_at</b>                      | timestamp            |                                      |
| <b>deleted_at</b>                      | timestamp<br>NULL    | soft delete                          |

| lessons (clases dictadas) |             |                      |
|---------------------------|-------------|----------------------|
| <b>id</b>                 | int PK      | AI                   |
| <b>coach_id</b>           | FK users.id | user con role_id = 2 |
| <b>level</b>              | varchar(50) | nivel                |
| <b>day_of_week</b>        | enum        | Mon..Sat             |
| <b>start_time</b>         | time        |                      |
| <b>end_time</b>           | time        |                      |
| <b>capacity</b>           | int         | cupos maximos        |
| <b>created_at</b>         | timestamp   |                      |

| password_resets   |              |        |
|-------------------|--------------|--------|
| <b>id</b>         | int PK       | AI     |
| <b>email</b>      | varchar(255) | indice |
| <b>token</b>      | varchar(255) | indice |
| <b>expires_at</b> | datetime     |        |
| <b>created_at</b> | timestamp    |        |

## Relaciones (ER simplificado)

| origen              | rel | destino           | comentario                 |
|---------------------|-----|-------------------|----------------------------|
| users.role_id       | N:1 | roles.id          | rol del usuario            |
| profiles.user_id    | 1:1 | users.id          | ON DELETE CASCADE          |
| lessons.coach_id    | N:1 | users.id (role=2) | validar rol al crear clase |
| bookings.swimmer_id | N:1 | users.id (role=3) | validar rol al inscribir   |
| bookings.lesson_id  | N:1 | lessons.id        | controlar capacity         |

### 3 Capa Models

Cada modelo encapsula UNA tabla. Los joins se hacen en el modelo dueño del listado (ejemplo: Lesson::getAllWithCoach()). Todos reciben el \$pdo por constructor (patron actual).

#### User.php

[EDITAR](#)

tabla: users

|                                               |                                       |
|-----------------------------------------------|---------------------------------------|
| <b>findByEmail(\$email)</b>                   | Login y validacion de existencia.     |
| <b>findById(\$id)</b>                         | Carga datos del usuario logueado.     |
| <b>create(\$email, \$hash, \$roleId)</b>      | Crear usuario; devuelve lastInsertId. |
| <b>updatePasswordById(\$id, \$hash)</b>       | Cambio de contraseña (panel).         |
| <b>updatePasswordByEmail(\$email, \$hash)</b> | Reset via token.                      |
| <b>countByRole(\$roleId)</b>                  | KPI para overview del admin.          |
| <b>getAllByRole(\$roleId)</b>                 | Listado por rol con JOIN profiles.    |
| <b>softDelete(\$id)</b>                       | deleted_at = NOW().                   |

#### Profile.php

[NUEVO](#)

tabla: profiles

|                                          |                                                                |
|------------------------------------------|----------------------------------------------------------------|
| <b>create(\$userId, \$data)</b>          | Crea perfil al registrar usuario.                              |
| <b>findById(\$userId)</b>                | Carga datos personales por user_id.                            |
| <b>updateByUserId(\$userId, \$data)</b>  | Edicion desde paneles coach/swimmer/admin.                     |
| <b>updateImage(\$userId, \$filename)</b> | Actualiza solo profile_image.                                  |
| <b>getCoachesList()</b>                  | Listar coaches con JOIN users (role=2) para selects de clases. |
| <b>getSwimmersList()</b>                 | Listado para overview admin.                                   |

**Lesson.php**

NUEVO

tabla: lessons

|                                                                    |                                                                         |
|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>getAllWithCoach()</b>                                           | Listado para admin con first_name+last_name del coach.                  |
| <b>getById(\$id)</b>                                               | Detalle (validar capacity, edit form).                                  |
| <b>getByCoachId(\$coachUserId)</b>                                 | Clases del profe logueado.                                              |
| <b>getAvailableForSwimmer(\$swimmerUserId)</b>                     | Oferta ordenada por dia/hora con cupos restantes y flag 'ya inscripto'. |
| <b>create(\$data) /<br/>update(\$id,\$data) /<br/>delete(\$id)</b> | CRUD admin.                                                             |

**Booking.php**

NUEVO

tabla: bookings

|                                                     |                                                              |
|-----------------------------------------------------|--------------------------------------------------------------|
| <b>create(\$swimmerUserId, \$lessonId)</b>          | Inserta status='Confirmed'.                                  |
| <b>cancel(\$bookingId, \$swimmerUserId)</b>         | status='Cancelled' (no DELETE).                              |
| <b>getBySwimmer(\$swimmerUserId)</b>                | Mis inscripciones.                                           |
| <b>countConfirmedByLesson(\$lessonId)</b>           | Validar capacity antes de crear.                             |
| <b>getStudentsByCoach(\$coachUserId, \$filters)</b> | JOIN bookings+lessons+users+profiles para listado del coach. |

**PasswordReset.php**

NUEVO

tabla: password\_resets

|                                              |                                       |
|----------------------------------------------|---------------------------------------|
| <b>create(\$email, \$token, \$expiresAt)</b> | Generar token al pedir reset.         |
| <b>findValidByToken(\$token)</b>             | Validar antes de cambiar password.    |
| <b>invalidate(\$token)</b>                   | Borrar / marcar usado tras el cambio. |

**Eliminados:** **Swimmer.php** y **Coach.php**. Sus responsabilidades pasan a **Profile.php** + filtrado por **role\_id** en **User.php**.

## 4 Capa Controllers

Cada controller hereda de BaseController. Usa `checkAuth($roleId)` para autorizacion y `render()` / `json()` para responder.

### LandingController.php

publico

NUEVO

`index()` Renderiza landing.view del club (sin DB).

### HomeController.php

todos

EDITAR

`index()` Switch por `role_id`; redirige a admin/coach/swimmer home.

### UserController.php

publico/auth

EDITAR

`login()` / `executeLogin()` Render + POST de login.

`register()` / `executeRegister()` Auto-registro de Swimmer (crea user + profile).

`forgotPassword()` / `executeForgot()` Genera token + envia email.

`resetPassword()` / `executeReset()` Valida token y actualiza password.

`logout()` Destruye sesion.

### AdminController.php

rol 1

EDITAR

`home()` Dashboard con KPIs (countByRole, etc).

`users()` Listado completo de usuarios.

`registerCoach()` / `executeRegistration()` Form + POST. **Genera password aleatoria** y dispara mail.

`coaches()` Listado de coaches (User+Profile, role=2).

`overview()` Vista de supervision general (todos los datos).

### LessonController.php

rol 1

NUEVO

`index()` Listado de clases con coach asociado.

`create()` / `store()` Form + POST nueva clase.

`edit($id)` / `update($id)` Form + POST editar clase.

`delete($id)` Eliminar clase (validar bookings asociados).

**CoachController.php**

rol 2

NUEVO

|                                                                           |                                                   |
|---------------------------------------------------------------------------|---------------------------------------------------|
| <b>home()</b>                                                             | Dashboard del profe.                              |
| <b>profile()</b> /<br><b>updateProfile()</b> /<br><b>updatePassword()</b> | Edicion de su perfil (Profile + User).            |
| <b>students()</b>                                                         | Listado de alumnos por dia/clase con filtros GET. |

**SwimmerController.php**

rol 3

NUEVO

|                                                                           |                                              |
|---------------------------------------------------------------------------|----------------------------------------------|
| <b>home()</b>                                                             | Dashboard del nadador.                       |
| <b>profile()</b> /<br><b>updateProfile()</b> /<br><b>updatePassword()</b> | Edicion de su perfil (Profile + User).       |
| <b>lessons()</b>                                                          | Oferta de clases ordenada por cronograma.    |
| <b>book()</b>                                                             | POST: valida cupo y duplicado; crea booking. |
| <b>myBookings()</b>                                                       | Listado de inscripciones del nadador.        |
| <b>cancelBooking(\$id)</b>                                                | POST: cambia status a Cancelled.             |

## 5 Capa Views

Las vistas se agrupan por rol. Cada rol con panel propio tiene su layout (header/footer). Las vistas siguen la convencion \*.view.php.

| app/views/(raiz)/              |                                            |        |
|--------------------------------|--------------------------------------------|--------|
| landing.view.php               | Landing institucional del club.            | NUEVO  |
| home.view.php                  | Vista basica/redirect (mantener fallback). | OK     |
| app/views/users/               |                                            |        |
| login.view.php                 | Form de login.                             | OK     |
| register.view.php              | Auto-registro de swimmer.                  | OK     |
| forgot-password.view.php       | Solicita reset.                            | OK     |
| reset-password.view.php        | Cambia password con token.                 | EDITAR |
| layout/header.php / footer.php | Layout publico/usuarios.                   | OK     |
| app/views/administrator/       |                                            |        |
| home.view.php                  | Dashboard admin con KPIs.                  | EDITAR |
| users.view.php                 | Listado general de usuarios.               | NUEVO  |
| coaches.view.php               | Listado coaches (User+Profile).            | EDITAR |
| register-coach.view.php        | Form alta coach (sin pedir password).      | EDITAR |
| lessons.view.php               | ABM de clases.                             | NUEVO  |
| lesson-form.view.php           | Form alta/edicion de clase.                | NUEVO  |
| overview.view.php              | Vista de supervision general.              | NUEVO  |
| app/views/coach/               |                                            |        |
| layout/header.php / footer.php | Layout exclusivo del profe.                | NUEVO  |
| home.view.php                  | Dashboard del profe.                       | NUEVO  |
| profile.view.php               | Datos personales + cambio password.        | NUEVO  |
| students.view.php              | Alumnos por dia/clase con filtros.         | NUEVO  |

| app/views/swimmer/             |                                     |       |
|--------------------------------|-------------------------------------|-------|
| layout/header.php / footer.php | Layout exclusivo del nadador.       | NUEVO |
| home.view.php                  | Dashboard del nadador.              | NUEVO |
| profile.view.php               | Datos personales + cambio password. | NUEVO |
| lessons.view.php               | Oferta de clases por cronograma.    | NUEVO |
| my-bookings.view.php           | Mis inscripciones (cancelar).       | NUEVO |

## 6 Rutas (router.php)

Convencion: ?route=<clave>. Las rutas estan agrupadas por modulo.

### Publico / Auth

| route    | metodo   | controller::action             | auth |
|----------|----------|--------------------------------|------|
| landing  | GET      | LandingController::index       | -    |
| login    | GET/POST | UserController::login          | -    |
| register | GET/POST | UserController::register       | -    |
| forgot   | GET/POST | UserController::forgotPassword | -    |
| reset    | GET/POST | UserController::resetPassword  | -    |
| logout   | GET      | UserController::logout         | auth |

### Administrador (rol 1)

| route          | metodo   | controller::action               | auth  |
|----------------|----------|----------------------------------|-------|
| admin-home     | GET      | AdminController::home            | rol 1 |
| admin-users    | GET      | AdminController::users           | rol 1 |
| admin-coaches  | GET      | AdminController::coaches         | rol 1 |
| coach-register | GET/POST | AdminController::registerCoach   | rol 1 |
| admin-overview | GET      | AdminController::overview        | rol 1 |
| lessons        | GET      | LessonController::index          | rol 1 |
| lesson-create  | GET/POST | LessonController::create / store | rol 1 |
| lesson-edit    | GET/POST | LessonController::edit / update  | rol 1 |
| lesson-delete  | POST     | LessonController::delete         | rol 1 |

**Coach (rol 2)**

| route          | metodo   | controller::action                       | auth  |
|----------------|----------|------------------------------------------|-------|
| coach-home     | GET      | CoachController::home                    | rol 2 |
| coach-profile  | GET/POST | CoachController::profile / updateProfile | rol 2 |
| coach-password | POST     | CoachController::updatePassword          | rol 2 |
| coach-students | GET      | CoachController::students                | rol 2 |

**Swimmer (rol 3)**

| route            | metodo   | controller::action                         | auth  |
|------------------|----------|--------------------------------------------|-------|
| swimmer-home     | GET      | SwimmerController::home                    | rol 3 |
| swimmer-profile  | GET/POST | SwimmerController::profile / updateProfile | rol 3 |
| swimmer-password | POST     | SwimmerController::updatePassword          | rol 3 |
| swimmer-lessons  | GET      | SwimmerController::lessons                 | rol 3 |
| swimmer-book     | POST     | SwimmerController::book                    | rol 3 |
| my-bookings      | GET      | SwimmerController::myBookings              | rol 3 |
| cancel-booking   | POST     | SwimmerController::cancelBooking           | rol 3 |

## 7 Mapeo Funcionalidad x Rol

| Funcionalidad                | Admin | Coach | Swimmer | Pieza clave                       |
|------------------------------|-------|-------|---------|-----------------------------------|
| Landing publica              | CRUD  | CRUD  | CRUD    | LandingController                 |
| Auth (login/registro/reset)  | CRUD  | CRUD  | CRUD    | UserController                    |
| Gestion usuarios             | CRUD  | -     | -       | AdminController + User+Profile    |
| Alta de profesores           | CRUD  | -     | -       | AdminController (mail provisorio) |
| Gestion de clases (CRUD)     | CRUD  | ver   | ver     | LessonController + Lesson         |
| Listado de alumnos por clase | CRUD  | CRUD  | -       | Booking::getStudentsByCoach       |
| Editar perfil propio         | CRUD  | CRUD  | CRUD    | Profile::updateById               |
| Cambiar password propia      | CRUD  | CRUD  | CRUD    | User::updatePasswordById          |
| Inscribirse a clases         | -     | -     | CRUD    | Booking::create                   |
| Cancelar inscripcion         | -     | -     | CRUD    | Booking::cancel                   |
| Supervision general          | CRUD  | -     | -       | AdminController::overview         |

**CRUD** = puede crear/editar/eliminar | **ver** = solo lectura | - = sin acceso.

## 8 Arbol final de archivos

Así queda el proyecto luego de aplicar la unificación. + NUEVO, ~ EDITAR, - ELIMINAR.

```
natacion-grupo2/
|-- public/
|   |-- index.php
|   |-- router.php           ~ agregar nuevas rutas
|-- app/
|   |-- config/
|   |   |-- db.php
|   |   |-- core/
|   |   |   |-- BaseController.php
|   |   |   |-- Env.php
|   |   |-- services/
|   |   |   |-- MailService.php           ~ + sendCoachWelcomeEmail()
|   |   |-- controllers/
|   |   |   |-- HomeController.php         ~ switch por role
|   |   |   |-- UserController.php        ~ usa Profile en register
|   |   |   |-- AdminController.php       ~ overview + mail provisorio
|   |   |   |-- LandingController.php     + NUEVO
|   |   |   |-- LessonController.php     + NUEVO
|   |   |   |-- CoachController.php      + NUEVO
|   |   |   |-- SwimmerController.php    + NUEVO
|   |   |-- models/
|   |   |   |-- User.php                 ~ countByRole / getAllByRole
|   |   |   |-- Profile.php              + NUEVO (reemplaza Swimmer/Coach)
|   |   |   |-- Lesson.php               + NUEVO
|   |   |   |-- Booking.php              + NUEVO
|   |   |   |-- PasswordReset.php        + NUEVO
|   |   |   |-- Swimmer.php              - ELIMINAR
|   |   |   |-- Coach.php                 - ELIMINAR
|   |   |-- views/
|   |   |   |-- landing.view.php          + NUEVO
|   |   |   |-- home.view.php
|   |   |   |-- users/
|   |   |   |   |-- login.view.php
|   |   |   |   |-- register.view.php
|   |   |   |   |-- forgot-password.view.php
|   |   |   |   |-- reset-password.view.php ~ usa token
|   |   |   |   |-- layout/{header,footer}.php
|   |   |   |-- administrator/
|   |   |   |   |-- home.view.php         ~ KPis
|   |   |   |   |-- users.view.php        + NUEVO
|   |   |   |   |-- coaches.view.php      ~ User+Profile
|   |   |   |   |-- register-coach.view.php ~ sin password input
|   |   |   |   |-- lessons.view.php      + NUEVO
|   |   |   |   |-- lesson-form.view.php  + NUEVO
|   |   |   |   |-- overview.view.php     + NUEVO
|   |   |   |-- coach/                   + NUEVO (carpeta)
|   |   |   |   |-- layout/{header,footer}.php
|   |   |   |   |-- home.view.php
|   |   |   |   |-- profile.view.php
|   |   |   |   |-- students.view.php
|   |   |   |-- swimmer/                 + NUEVO (carpeta)
|   |   |   |   |-- layout/{header,footer}.php
|   |   |   |   |-- home.view.php
|   |   |   |   |-- profile.view.php
|   |   |   |   |-- lessons.view.php
|   |   |   |-- my-bookings.view.php
```

## 9 Checklist de migracion

### DB - migrar datos

- Crear migracion SQL: copiar registros de swimmers y coaches a profiles (uniendo first\_name, last\_name, phone, birth\_date, profile\_image, specialty).
- Modificar lessons.coach\_id: pasar a referenciar users.id (en vez de coaches.id).
- Modificar bookings.swimmer\_id: pasar a referenciar users.id (en vez de swimmers.id).
- Eliminar tablas swimmers y coaches una vez migrados los FK.
- Verificar UNIQUE(swimmer\_id, lesson\_id) en bookings.

### Codigo - models

- Crear Profile.php con metodos definidos en seccion 3.
- Crear Lesson.php, Booking.php, PasswordReset.php.
- Eliminar Swimmer.php y Coach.php; reemplazar usos por Profile + User.
- Ajustar User.php para countByRole / getAllByRole.

### Codigo - controllers

- Crear LandingController, LessonController, CoachController, SwimmerController.
- Editar AdminController: generar password aleatoria al crear coach + mail.
- Editar UserController.register: crear user + profile (transaccion).
- Editar HomeController: ramificar al home de cada rol.

### Codigo - views

- Crear vistas de coach/ y swimmer/ con su layout.
- Crear administrator/lessons.view.php, lesson-form.view.php, overview.view.php.
- Quitar inputs de password en register-coach.view.php.

### Servicios

- MailService::sendCoachWelcomeEmail (template HTML con password provisoria).
- MailService::sendPasswordResetEmail (si no esta).

### Validaciones

- Inscripcion: validar capacity > count(Confirmed) y no duplicados.
- Crear/editar clase: validar coach\_id pertenece a un user con role\_id=2.
- Cambio de password: pedir password actual y validar largo minimo.

---

Una vez completados los pasos, la app deberia funcionar exactamente igual que con swimmers/coaches separados, pero con un modelo de datos mas simple y consistente.